

# Cisco IOS-XE

# Network Address Translation

## Comprehensive Lab Manual

*Inside Static NAT · Outside Static NAT · Dynamic NAT · PAT*

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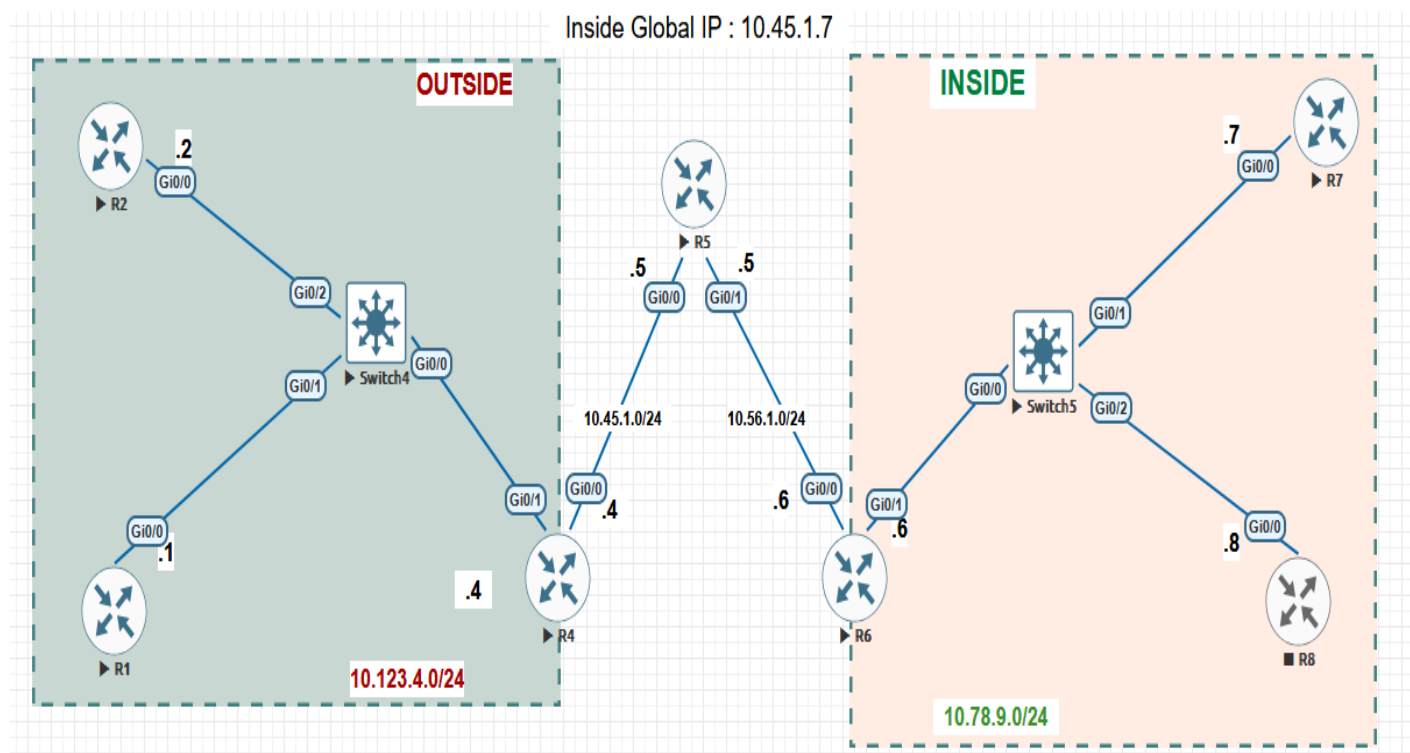
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## Lab Topology & Addressing



## Overview

This lab uses a two-zone topology separated by R5, which performs all NAT operations. The **OUTSIDE** zone contains R1, R2, R4 and Switch4; the **INSIDE** zone contains R6, R7, R8, and Switch5. R4 and R6 serve as aggregation routers connecting to R5.

## Network Zones

| Zone                 | Subnet        | Devices                      | Description                            |
|----------------------|---------------|------------------------------|----------------------------------------|
| OUTSIDE              | 10.123.4.0/24 | R1, R2, Switch4, R4<br>Gi0/1 | Simulates the external internet        |
| TRANSIT<br>(Outside) | 10.45.1.0/24  | R5 Gi0/0, R4 Gi0/0           | WAN link — Outside NAT interface       |
| TRANSIT<br>(Inside)  | 10.56.1.0/24  | R5 Gi0/1, R6 Gi0/1           | WAN link — Inside NAT interface        |
| INSIDE               | 10.78.9.0/24  | R7, R8, Switch5, R6<br>Gi0/0 | Simulates the private internal network |

## IP Addressing Table

| Device | Interface | IP Address    | Role                  |
|--------|-----------|---------------|-----------------------|
| R1     | Gi0/0     | 10.123.4.1/24 | Outside client/server |
| R2     | Gi0/0     | 10.123.4.2/24 | Outside client/server |
| R4     | Gi0/1     | 10.123.4.4/24 | Outside aggregation   |
| R4     | Gi0/0     | 10.45.1.4/24  | R4↔R5 link            |
| R5     | Gi0/0     | 10.45.1.5/24  | NAT OUTSIDE interface |
| R5     | Gi0/1     | 10.56.1.5/24  | NAT INSIDE interface  |
| R6     | Gi0/1     | 10.56.1.6/24  | R5↔R6 link            |
| R6     | Gi0/0     | 10.78.9.6/24  | Inside aggregation    |
| R7     | Gi0/0     | 10.78.9.7/24  | Inside client/server  |
| R8     | Gi0/0     | 10.78.9.8/24  | Inside client/server  |

## Static Routing Summary

| Router | Command                                     | Meaning            |
|--------|---------------------------------------------|--------------------|
| R1     | ip route 0.0.0.0 0.0.0.0 10.123.4.4         | Default → R4       |
| R2     | ip route 0.0.0.0 0.0.0.0 10.123.4.4         | Default → R4       |
| R4     | ip route 0.0.0.0 0.0.0.0 10.45.1.5          | Default → R5       |
| R5     | ip route 10.123.4.0 255.255.255.0 10.45.1.4 | Outside LAN via R4 |
| R5     | ip route 10.78.9.0 255.255.255.0 10.56.1.6  | Inside LAN via R6  |
| R6     | ip route 0.0.0.0 0.0.0.0 10.56.1.5          | Default → R5       |
| R7     | ip route 0.0.0.0 0.0.0.0 10.78.9.6          | Default → R6       |
| R8     | ip route 0.0.0.0 0.0.0.0 10.78.9.6          | Default → R6       |

### NAT Terminology Recap

Inside Local — the real private IP of an inside host (e.g. 10.78.9.7).

Inside Global — how an inside host appears to the outside (translated IP, e.g. 10.45.1.7).

Outside Local — how an outside host appears to the inside (e.g. 10.123.4.222).

Outside Global — the real public IP of an outside host (e.g. 10.123.4.2).

All translations are performed exclusively on R5.

## Base Configuration (All Labs)

Apply these configurations once before starting any individual lab scenario. Every subsequent scenario builds on this foundation.

### R1 — Outside Client

#### IOS-XE Configuration

```
hostname R1
!
interface GigabitEthernet0/0
 ip address 10.123.4.1 255.255.255.0 ! Outside LAN
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.123.4.4 ! Default route to R4
```

### R2 — Outside Client

#### IOS-XE Configuration

```
hostname R2
!
interface GigabitEthernet0/0
 ip address 10.123.4.2 255.255.255.0 ! Outside LAN
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.123.4.4 ! Default route to R4
```

### R4 — Outside Aggregation

#### IOS-XE Configuration

```
hostname R4
!
interface GigabitEthernet0/0
 ip address 10.45.1.4 255.255.255.0 ! Link to R5 Gi0/0 (transit)
 no shutdown
!
interface GigabitEthernet0/1
 ip address 10.123.4.4 255.255.255.0 ! Outside LAN (Switch4 side)
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.45.1.5 ! Default route to R5
```

### R5 — NAT Router (Base Interfaces & Routes)

#### ⚠ Reset Before Each Scenario

Before starting each lab scenario on R5, remove all NAT statements and ip nat commands from the previous scenario. The base interface and routing config below remains constant.

#### IOS-XE Configuration

```
hostname R5
!
interface GigabitEthernet0/0
 ip address 10.45.1.5 255.255.255.0 ! OUTSIDE NAT interface (added per scenario)
```

```
no shutdown
!
interface GigabitEthernet0/1
 ip address 10.56.1.5 255.255.255.0 ! INSIDE NAT interface (added per scenario)
 no shutdown
!
ip route 10.123.4.0 255.255.255.0 10.45.1.4 ! Reach the outside LAN via R4
ip route 10.78.9.0 255.255.255.0 10.56.1.6 ! Reach the inside LAN via R6
```

## R6 — Inside Aggregation

### IOS-XE Configuration

```
hostname R6
!
interface GigabitEthernet0/0
 ip address 10.78.9.6 255.255.255.0 ! Inside LAN (Switch5 side)
 no shutdown
!
interface GigabitEthernet0/1
 ip address 10.56.1.6 255.255.255.0 ! Link to R5 Gi0/1 (transit)
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.56.1.5 ! Default route to R5
```

## R7 — Inside Client

### IOS-XE Configuration

```
hostname R7
!
interface GigabitEthernet0/0
 ip address 10.78.9.7 255.255.255.0 ! Inside LAN
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.78.9.6 ! Default route to R6
```

## R8 — Inside Client

### IOS-XE Configuration

```
hostname R8
!
interface GigabitEthernet0/0
 ip address 10.78.9.8 255.255.255.0 ! Inside LAN
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.78.9.6 ! Default route to R6
```

### ☒ Verify Reachability Before NAT

Before configuring NAT, verify end-to-end IP reachability using the routing table only:

```
R5# ping 10.123.4.1 source 10.56.1.5 (should FAIL without NAT — that is expected)
R5# ping 10.45.1.4 (should succeed — direct link)
R5# ping 10.56.1.6 (should succeed — direct link)
```

## Lab 1 — Inside Static NAT

### Objective

Configure a one-to-one static translation so that inside host R7 (10.78.9.7) is permanently mapped to the Inside Global IP 10.45.1.7. Any packet from the outside addressed to 10.45.1.7 will be forwarded to R7, and any packet sourced by R7 will appear to come from 10.45.1.7.

### Concept

Inside Static NAT creates a permanent, bidirectional mapping between one Inside Local address and one Inside Global address. It is commonly used for servers that must be reachable from the outside using a stable routable address.

| NAT Term       | IP Address | Notes                                                                    |
|----------------|------------|--------------------------------------------------------------------------|
| Inside Local   | 10.78.9.7  | R7's real IP — exists only inside the private network                    |
| Inside Global  | 10.45.1.7  | How R7 appears to outside hosts — routable on the internet-facing subnet |
| Outside Local  | 10.123.4.1 | How R1 appears to inside hosts (unchanged in Inside Static NAT)          |
| Outside Global | 10.123.4.1 | R1's real IP (same as Outside Local for this scenario)                   |

### Packet Walk-Through

Outbound (R7 → R1):

- R7 sends a packet: Src=10.78.9.7, Dst=10.123.4.1
- Packet arrives on R5 Gi0/1 (inside interface).
- R5 matches the inside local address 10.78.9.7 against the static NAT entry.
- R5 replaces the source IP: Src becomes 10.45.1.7, Dst remains 10.123.4.1.
- Translated packet exits R5 Gi0/0 toward R4 → R1.

Inbound (R1 → R7):

- R1 sends a packet: Src=10.123.4.1, Dst=10.45.1.7.
- Packet arrives on R5 Gi0/0 (outside interface).
- R5 matches destination 10.45.1.7 against the static NAT table.
- R5 replaces the destination IP: Dst becomes 10.78.9.7.
- Translated packet exits R5 Gi0/1 toward R6 → R7.

### R5 Configuration

#### Clean Slate

Ensure no previous NAT configuration exists on R5. Run: R5# show run | include ip nat — remove any leftover entries before proceeding.



**IOS-XE Configuration**

```
interface GigabitEthernet0/0 ! Outside interface
 ip nat outside ! Mark as NAT outside
!
interface GigabitEthernet0/1 ! Inside interface
 ip nat inside ! Mark as NAT inside
!
ip nat inside source static 10.78.9.7 10.45.1.7 ! Inside Local → Inside Global
(permanent)
```

## Verification

### Step 1 — Inspect the NAT Table

**IOS-XE Configuration**

```
R5# show ip nat translations
```

Expected output before any traffic:

**IOS-XE Configuration**

| Pro | Inside global | Inside local | Outside local | Outside global |
|-----|---------------|--------------|---------------|----------------|
| --- | 10.45.1.7     | 10.78.9.7    | ---           | ---            |

### Step 2 — Generate Traffic from R7

**IOS-XE Configuration**

```
R7# ping 10.123.4.1 repeat 5
R7# ping 10.123.4.2 repeat 5
```

### Step 3 — Inspect the NAT Table After Traffic

**IOS-XE Configuration**

```
R5# show ip nat translations
```

Expected output (ICMP entry added dynamically):

**IOS-XE Configuration**

| Pro  | Inside global | Inside local | Outside local | Outside global |
|------|---------------|--------------|---------------|----------------|
| icmp | 10.45.1.7:1   | 10.78.9.7:1  | 10.123.4.1:1  | 10.123.4.1:1   |
| ---  | 10.45.1.7     | 10.78.9.7    | ---           | ---            |

### Step 4 — Generate Traffic from R1 to the Inside Global IP

**IOS-XE Configuration**

```
R1# ping 10.45.1.7 repeat 5
```

**☑ What to Expect**

R1's ping to 10.45.1.7 is forwarded by R4 to R5.  
R5 translates Dst 10.45.1.7 → 10.78.9.7 and forwards to R7.  
R7 replies; R5 translates Src 10.78.9.7 → 10.45.1.7 back to R1.  
Result: R1 sees replies from 10.45.1.7 — it never learns R7's real address.

### Step 5 — Statistics

**IOS-XE Configuration**

```
R5# show ip nat statistics
```

Key fields to check: 'Total active translations', 'Outside interfaces', 'Inside interfaces', 'Hits' and 'Misses' counters.

## Step 6 — Debug (Optional)

### IOS-XE Configuration

```
R5# debug ip nat
R7# ping 10.123.4.1
R5# undebug all
```

### ⚠ Debug in Production

Never run 'debug ip nat' on a production router with high traffic — it can saturate the CPU.

## Lab 2 — Outside Static NAT

### Objective

Configure a static translation so that outside host R2 (10.123.4.2 — the Outside Global) appears to inside hosts as 10.123.4.222 (the Outside Local). This allows policy or access control on the inside to reference a predictable address for an outside resource, regardless of its real IP.

### Concept

Outside Static NAT translates the destination address of packets coming from the outside, and the source address of return packets going back to the outside. Inside hosts contact 10.123.4.222 and R5 translates that to/from the real address 10.123.4.2. This is the mirror image of Inside Static NAT.

| NAT Term       | IP Address   | Notes                                                   |
|----------------|--------------|---------------------------------------------------------|
| Inside Local   | 10.78.9.7    | R7's real IP (unchanged in Outside Static NAT)          |
| Inside Global  | 10.78.9.7    | Same — inside addresses not translated in this scenario |
| Outside Local  | 10.123.4.222 | How R2 appears to the inside hosts — the 'fake' address |
| Outside Global | 10.123.4.2   | R2's real IP on the outside network                     |

### Packet Walk-Through

Outbound (R7 → R2 using Outside Local):

- R7 sends a packet: Src=10.78.9.7, Dst=10.123.4.222 (the outside local address).
- Packet arrives on R5 Gi0/1 (inside interface).
- R5 matches Dst 10.123.4.222 — the outside local address — in its NAT table.
- R5 replaces the destination: Dst becomes 10.123.4.2 (the real IP of R2).
- Source address 10.78.9.7 is NOT translated (only outside address is mapped here).
- Packet exits R5 Gi0/0 toward R4 → R2.

Inbound (R2 → R7, return traffic):

- R2 sends a reply: Src=10.123.4.2, Dst=10.78.9.7.
- Packet arrives on R5 Gi0/0 (outside interface).
- R5 matches Src 10.123.4.2 and translates it to 10.123.4.222 for inside hosts.
- Packet exits R5 Gi0/1 toward R6 → R7.
- R7 sees the packet as coming from 10.123.4.222, confirming the outside local mapping.

### R5 Configuration

#### ⚠ Remove Previous NAT Config

Before applying this config, remove the Inside Static NAT entry from Lab 1:

```
R5(config)# no ip nat inside source static 10.78.9.7 10.45.1.7
```

```
R5(config)# interface Gi0/0 → no ip nat outside
```

```
R5(config)# interface Gi0/1 → no ip nat inside
```

**IOS-XE Configuration**

```
interface GigabitEthernet0/0 ! Outside interface
ip nat outside ! Mark as NAT outside
!
interface GigabitEthernet0/1 ! Inside interface
ip nat inside ! Mark as NAT inside
!
ip nat outside source static 10.123.4.2 10.123.4.222 ! Outside Global → Outside
Local (R2's real IP → fake inside-facing IP)
```

 **Command Syntax Distinction**

'ip nat inside source static' translates INSIDE addresses.

'ip nat outside source static' translates OUTSIDE addresses.

The keyword 'outside' after 'nat' tells IOS-XE which pool of addresses is being aliased.

## R7 — Add a Route to the Outside Local Address

R7 must be able to reach 10.123.4.222 (the outside local address). Since 10.123.4.222 is not in R7's directly connected subnet, it uses its default route toward R6 → R5, which handles the translation. No extra route is needed on R7 as long as the default route exists.

 **Why No Extra Route?**

R7 sends traffic to 10.123.4.222 via its default route (→ R6 → R5).

R5 intercepts the packet on Gi0/1 and translates 10.123.4.222 to 10.123.4.2.

R5 then routes toward R4 → R2 using its static route to 10.123.4.0/24.

## Verification

### Step 1 — Inspect the NAT Table

**IOS-XE Configuration**

```
R5# show ip nat translations
```

Expected output before traffic:

**IOS-XE Configuration**

| Pro | Inside global | Inside local | Outside local | Outside global |
|-----|---------------|--------------|---------------|----------------|
| --- | ---           | ---          | 10.123.4.222  | 10.123.4.2     |

### Step 2 — Generate Traffic from R7 to the Outside Local Address

**IOS-XE Configuration**

```
R7# ping 10.123.4.222 repeat 5
```

R7 targets 10.123.4.222. R5 translates destination to 10.123.4.2 and forwards to R2.

### Step 3 — Inspect the NAT Table After Traffic

**IOS-XE Configuration**

```
R5# show ip nat translations verbose
```

Expected output:

**IOS-XE Configuration**

| Pro  | Inside global | Inside local | Outside local  | Outside global |
|------|---------------|--------------|----------------|----------------|
| icmp | 10.78.9.7:1   | 10.78.9.7:1  | 10.123.4.222:1 | 10.123.4.2:1   |
| ---  | ---           | ---          | 10.123.4.222   | 10.123.4.2     |

## Step 4 — Confirm from R2's Perspective

**IOS-XE Configuration**

```
R2# ping 10.78.9.7 repeat 5
```

R2 pings R7 directly. R5 translates the source from R2's real IP (10.123.4.2) to the outside local (10.123.4.222) as the packet enters the inside.

## Lab 3 — Dynamic NAT (Pool-Based)

### Objective

Configure Dynamic NAT using a pool of two addresses (10.45.1.10 and 10.45.1.11) so that inside hosts R7 and R8 are dynamically mapped to pool addresses on a first-come, first-served basis. Translations are created on demand and removed after a configurable timeout.

### Concept

Dynamic NAT maps inside local addresses to a pool of inside global addresses. When an inside host initiates a connection, R5 selects the next available address from the pool and creates a temporary translation entry. If the pool is exhausted, new connections are dropped until an entry times out.

| NAT Term       | IP Address / Range      | Notes                                             |
|----------------|-------------------------|---------------------------------------------------|
| Inside Local   | 10.78.9.7, 10.78.9.8    | R7 and R8's real private IPs                      |
| Inside Global  | 10.45.1.10 – 10.45.1.11 | Pool — assigned dynamically by R5                 |
| Outside Local  | 10.123.4.x              | Outside hosts' addresses — not translated here    |
| Outside Global | 10.123.4.x              | Same — outside addresses unchanged in Dynamic NAT |

#### Pool Size Limitation

This pool contains only TWO addresses (10.45.1.10 and 10.45.1.11).

If a third inside host initiates a connection simultaneously, it will be dropped.

PAT (Lab 4) overcomes this limitation by sharing a single address across thousands of sessions.

### Packet Walk-Through

Outbound (R7 → R1, first host):

- R7 sends: Src=10.78.9.7, Dst=10.123.4.1.
- R5 checks the access-list — 10.78.9.7 is permitted.
- R5 picks the first free pool address: 10.45.1.10.
- Translation entry created: 10.78.9.7 ↔ 10.45.1.10.
- Packet exits with Src=10.45.1.10.

Outbound (R8 → R1, second host):

- R8 sends: Src=10.78.9.8, Dst=10.123.4.1.
- R5 picks the next free pool address: 10.45.1.11.
- Translation entry created: 10.78.9.8 ↔ 10.45.1.11.
- Packet exits with Src=10.45.1.11.

### R5 Configuration

### ⚠ Remove Previous NAT Config First

```
R5(config)# no ip nat outside source static 10.123.4.2 10.123.4.222
R5(config)# interface Gi0/0 → no ip nat outside
R5(config)# interface Gi0/1 → no ip nat inside
```

#### IOS-XE Configuration

```
interface GigabitEthernet0/0 ! Outside interface
 ip nat outside
!
interface GigabitEthernet0/1 ! Inside interface
 ip nat inside
!
ip access-list standard NAT_INSIDE_HOSTS ! Define which inside hosts are translated
 permit 10.78.9.0 0.0.0.255 ! Permit entire inside LAN
!
ip nat pool DYNAMIC_POOL 10.45.1.10 10.45.1.11 netmask 255.255.255.0 ! Pool of
two inside global addresses
!
ip nat inside source list NAT_INSIDE_HOSTS pool DYNAMIC_POOL ! Bind ACL to pool —
dynamic NAT
```

### 📄 Pool Netmask

The netmask in 'ip nat pool' defines the subnet the pool belongs to, NOT the range size. 10.45.1.10 to 10.45.1.11 with /24 mask means both IPs live on the 10.45.1.0/24 subnet. The range is set by the start and end IP parameters — only two addresses are usable here.

## Verification

### Step 1 — Generate Traffic from R7 and R8 Simultaneously

#### IOS-XE Configuration

```
R7# ping 10.123.4.1 repeat 10
R8# ping 10.123.4.1 repeat 10
```

### Step 2 — View Dynamic Translations

#### IOS-XE Configuration

```
R5# show ip nat translations
```

Expected output (both hosts actively translating):

#### IOS-XE Configuration

| Pro  | Inside global | Inside local | Outside local | Outside global |
|------|---------------|--------------|---------------|----------------|
| icmp | 10.45.1.10:1  | 10.78.9.7:1  | 10.123.4.1:1  | 10.123.4.1:1   |
| icmp | 10.45.1.11:1  | 10.78.9.8:1  | 10.123.4.1:1  | 10.123.4.1:1   |

### Step 3 — View Statistics

#### IOS-XE Configuration

```
R5# show ip nat statistics
```

Key fields: 'Pool stats' → 'Allocated' should show 2 out of 2; 'Misses' should be 0 if traffic was sequential.

## Step 4 — Pool Exhaustion Test (Optional)

Initiate a third simultaneous translation attempt by pinging from an additional host (VPC9). Observe the 'Misses' counter increment on 'show ip nat statistics'. Or 'debug ip nat detailed' and observe 'NAT: failed to allocate address for 10.78.9.9'. 'no debug ip nat detailed' to stop debugging.

## Step 5 — Observe Timeout Behavior

IOS-XE Configuration

```
R5# show ip nat translations verbose
```

The 'expires in' field shows the remaining time for each dynamic entry. ICMP entries default to 60 seconds; TCP/UDP entries default to 300/86400 seconds respectively.

## Step 6 — Clear Translations Manually

IOS-XE Configuration

```
R5# clear ip nat translation * ! Removes all dynamic entries; static entries remain
```



## Lab 4 — PAT (Port Address Translation / NAT Overload)

### Objective

Configure PAT so that all inside hosts (R7 and R8) share a single outside IP address (R5's Gi0/0 address, 10.45.1.5) by differentiating sessions using source port numbers. This is the most common NAT type used in enterprise and home networks.

### Concept

PAT — also called NAT Overload — extends Dynamic NAT by adding the source port as an additional dimension for translation. Thousands of inside hosts can share one public IP because each session is uniquely identified by the combination of IP address + port number. R5 maintains a port mapping table in addition to the address translation table.

| NAT Term       | Value                             | Notes                                    |
|----------------|-----------------------------------|------------------------------------------|
| Inside Local   | 10.78.9.7:1024,<br>10.78.9.8:1025 | R7 and R8 with unique source ports       |
| Inside Global  | 10.45.1.5:2049,<br>10.45.1.5:2050 | Same IP, different ports — PAT in action |
| Outside Local  | 10.123.4.1                        | Outside host — address not translated    |
| Outside Global | 10.123.4.1                        | Same as Outside Local                    |

### Packet Walk-Through

Outbound (R7, port 1024 → R1, port 23):

- R7 sends: Src=10.78.9.7:1024, Dst=10.123.4.1:23.
- R5 matches the ACL and selects the overload address (Gi0/0 = 10.45.1.5).
- R5 assigns a unique port: 10.45.1.5:2049.
- PAT entry created: 10.78.9.7:1024 ↔ 10.45.1.5:2049.
- Packet exits: Src=10.45.1.5:2049, Dst=10.123.4.1:23.

Outbound (R8, port 1025 → R1, port 23 — same time):

- R8 sends: Src=10.78.9.8:1025, Dst=10.123.4.1:23.
- R5 assigns a different port: 10.45.1.5:2050.
- PAT entry created: 10.78.9.8:1025 ↔ 10.45.1.5:2050.
- Packet exits: Src=10.45.1.5:2050, Dst=10.123.4.1:23.

Inbound (Reply from R1 to R7):

- R1 sends: Src=10.123.4.1:23, Dst=10.45.1.5:2049.
- R5 looks up port 2049 in the PAT table → matches R7:1024.
- R5 translates Dst to 10.78.9.7:1024 and forwards to R7.

## R5 Configuration — PAT Using Interface Address

### ⚠ Remove Previous NAT Config First

```
R5(config)# no ip nat inside source list NAT_INSIDE_HOSTS pool DYNAMIC_POOL
R5(config)# no ip nat pool DYNAMIC_POOL 10.45.1.10 10.45.1.11 netmask 255.255.255.0
R5(config)# interface Gi0/0 → no ip nat outside
R5(config)# interface Gi0/1 → no ip nat inside
Keep the NAT_INSIDE_HOSTS ACL — it is reused below.
```

#### IOS-XE Configuration

```
interface GigabitEthernet0/0 ! Outside interface
 ip nat outside
!
interface GigabitEthernet0/1 ! Inside interface
 ip nat inside
!
ip access-list standard NAT_INSIDE_HOSTS ! Reuse or recreate if needed
 permit 10.78.9.0 0.0.0.255
!
ip nat inside source list NAT_INSIDE_HOSTS interface GigabitEthernet0/0
 overload ! PAT — use Gi0/0's IP (10.45.1.5), overload = enable port multiplexing
```

### 📄 The 'overload' Keyword

Adding 'overload' to the 'ip nat inside source' command enables PAT. Without 'overload' + 'interface', it would behave as Dynamic NAT (requiring a pool). Using 'interface GigabitEthernet0/0' means R5 automatically uses whatever IP is on that interface — no pool needed.

## Alternative: PAT Using a Pool

PAT can also be configured with a pool if you want to use a specific address or range rather than the interface IP:

#### IOS-XE Configuration

```
ip nat pool PAT_POOL 10.45.1.10 10.45.1.10 netmask 255.255.255.0 ! Pool with a
single IP for PAT
ip nat inside source list NAT_INSIDE_HOSTS pool PAT_POOL overload ! Note
'overload' at the end — same keyword enables PAT
```

## Verification

### Step 1 — Generate Traffic from Both Inside Hosts

#### IOS-XE Configuration

```
R7# ping 10.123.4.1 repeat 10
R8# ping 10.123.4.2 repeat 10
```

### Step 2 — View PAT Translations

#### IOS-XE Configuration

```
R5# show ip nat translations
```

Expected output (port numbers visible):

**IOS-XE Configuration**

| Pro  | Inside global | Inside local | Outside local | Outside global |
|------|---------------|--------------|---------------|----------------|
| icmp | 10.45.1.5:1   | 10.78.9.7:1  | 10.123.4.1:1  | 10.123.4.1:1   |
| icmp | 10.45.1.5:2   | 10.78.9.8:2  | 10.123.4.2:2  | 10.123.4.2:2   |

Notice both inside hosts share the same Inside Global IP (10.45.1.5) but use different port numbers.

### Step 3 — Statistics

**IOS-XE Configuration**

```
R5# show ip nat statistics
```

'Total translations' will show entries; 'Hits' should increase with each translated packet. No pool exhaustion is possible since ports can range from 1024 to 65535.

### Step 4 — Stress Test PAT

**IOS-XE Configuration**

```
R7# ping 10.123.4.2 repeat 100
R8# ping 10.123.4.1 repeat 100
R5# show ip nat translations | count 10.45.1.5
```

#### ☒ PAT Scalability

A single PAT address supports up to ~64,000 simultaneous sessions (ports 1024–65535). This is why PAT is used in virtually every home router and most enterprise environments. IOS-XE will display protocol and port in the NAT translation table for PAT entries.

# Troubleshooting NAT

## Essential Show Commands

| Command                          | What It Shows                                           |
|----------------------------------|---------------------------------------------------------|
| show ip nat translations         | All active translation entries (static + dynamic + PAT) |
| show ip nat translations verbose | Adds expiry timers and flags to each entry              |
| show ip nat statistics           | Hit/miss counters, pool usage, interface assignment     |
| show ip nat translations total   | Count of active translations                            |
| show run   include ip nat        | All NAT configuration lines in the running config       |
| show run   section ip nat        | Larger context around NAT config                        |
| show access-lists                | ACL hit counters — confirms ACL is matching traffic     |

## Debug Commands

### ⚠ Use Debugs with Caution

These commands generate significant output. Always disable with 'undebug all' when done.

#### IOS-XE Configuration

```
R5# debug ip nat ! Shows each packet being translated (high volume)
R5# debug ip nat detailed ! Includes packet contents and decision path
R5# undebug all ! Disable all debugs
```

## Common Issues & Fixes

| Symptom                                           | Likely Cause                               | Fix                                                                              |
|---------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------|
| NAT table empty after traffic                     | Interface not marked ip nat inside/outside | Add 'ip nat inside' and 'ip nat outside' to correct interfaces                   |
| Translations created but ping fails               | Missing return route on R5                 | Verify 'ip route 10.123.4.0 255.255.255.0 10.45.1.4' and '10.78.9.0' route exist |
| 'Misses' counter high                             | ACL not matching inside hosts              | Check ACL with 'show access-lists'; ensure permit covers the inside LAN          |
| Dynamic pool exhausted                            | Pool too small for concurrent hosts        | Expand pool range or switch to PAT with 'overload'                               |
| PAT: Inside hosts can't reach outside             | Wrong interface in overload command        | Ensure 'interface GigabitEthernet0/0' matches the actual outside interface       |
| Outside static: return traffic fails              | R7 routing to wrong next-hop               | Confirm R7's default route points to R6, not directly to R5                      |
| Static NAT: Translation appears but traffic fails | Routing loop or missing route              | Trace with 'debug ip packet' to identify the hop where traffic is dropped        |

## Clearing NAT State

### IOS-XE Configuration

```
R5# clear ip nat translation * ! Clear ALL dynamic translations (static entries remain)
R5# clear ip nat translation inside 10.78.9.7 10.45.1.7 ! Clear a specific translation
R5# clear ip nat translation tcp inside 10.78.9.7 80 outside 10.123.4.1 1234 ! Clear specific TCP entry
```

## NAT Translation Timeouts (Defaults)

| Protocol / Type | Default Timeout        | Adjust Command                            |
|-----------------|------------------------|-------------------------------------------|
| ICMP            | 60 seconds             | ip nat translation icmp-timeout <seconds> |
| UDP             | 300 seconds            | ip nat translation udp-timeout <seconds>  |
| TCP             | 86400 seconds (24 hrs) | ip nat translation tcp-timeout <seconds>  |
| TCP ESTABLISHED | 86400 seconds          | ip nat translation tcp-timeout <seconds>  |
| Static          | Never expires          | N/A — static entries are permanent        |

## Quick Reference Summary

### NAT Type Comparison

| NAT Type       | Command Keyword                                          | Addresses Used             | Bidirectional?    | Use Case                               |
|----------------|----------------------------------------------------------|----------------------------|-------------------|----------------------------------------|
| Inside Static  | ip nat inside source static <local> <global>             | 1-to-1 permanent           | Yes               | Servers needing a fixed outside IP     |
| Outside Static | ip nat outside source static <global> <local>            | 1-to-1 permanent (outside) | Yes               | Fixed inside alias for an outside host |
| Dynamic NAT    | ip nat inside source list <acl> pool <pool>              | Pool of IPs                | Only while active | Groups of hosts, limited pool          |
| PAT            | ip nat inside source list <acl> interface <int> overload | Single IP + ports          | Only while active | Many hosts, single public IP           |

### Key IOS-XE Syntax Cheat Sheet

#### IOS-XE Configuration

```

! Mark interfaces
interface Gi0/0 → ip nat outside
interface Gi0/1 → ip nat inside

! Inside Static NAT
ip nat inside source static <inside-local> <inside-global>

! Outside Static NAT
ip nat outside source static <outside-global> <outside-local>

! Dynamic NAT
ip nat pool <name> <start-ip> <end-ip> netmask <mask>
ip nat inside source list <acl> pool <name>

! PAT (interface)
ip nat inside source list <acl> interface <outside-int> overload

! PAT (pool)
ip nat inside source list <acl> pool <name> overload

! Verification
show ip nat translations
show ip nat translations verbose
show ip nat statistics
clear ip nat translation *
debug ip nat

```

### Address Mapping Summary for This Lab

| Lab                    | Inside Local             | Inside Global           | Outside Local          | Outside Global |
|------------------------|--------------------------|-------------------------|------------------------|----------------|
| Lab 1 — Inside Static  | 10.78.9.7                | 10.45.1.7               | 10.123.4.1 (no change) | 10.123.4.1     |
| Lab 2 — Outside Static | 10.78.9.7 (no change)    | 10.78.9.7               | 10.123.4.222           | 10.123.4.2     |
| Lab 3 — Dynamic NAT    | 10.78.9.7 / .8           | 10.45.1.10 / .11        | 10.123.4.x (no change) | 10.123.4.x     |
| Lab 4 — PAT            | 10.78.9.7:port / .8:port | 10.45.1.5:port (shared) | 10.123.4.x (no change) | 10.123.4.x     |

### Lab Completion Checklist

- ☐ Lab 1: R7 (10.78.9.7) translates to 10.45.1.7 — R1 can ping 10.45.1.7 and reach R7.
- ☐ Lab 2: R2 (10.123.4.2) appears as 10.123.4.222 to R7 — R7 pings 10.123.4.222 successfully.
- ☐ Lab 3: R7 and R8 each receive a unique pool address (10.45.1.10 or .11) — confirmed in NAT table.
- ☐ Lab 4: R7 and R8 both share 10.45.1.5 with different port numbers — confirmed in NAT table.
- ☐ All labs: 'show ip nat statistics' shows 0 misses and non-zero hits after traffic generation.